Criptografia Simétrica e Modos de Operação

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Gerando uma chave simétrica

pera@Pera: ~/facul/seguranca/Blocos

```
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -pbkdf2 -aes-128-ecb -k my-secret-passphrase -P
salt=A7594BD44D934309
key=7A985394F8301A21F17F153EF492AB00
pera@Pera:~/facul/seguranca/Blocos$ echo 7A985394F8301A21F17F153EF492AB00 > key.priv
```

Gerei a chave de 128 bits a partir de uma senha com o PBKDF2 e salvei em `key.priv`.

Criptografia com AES-128

pera@Pera: ~/facul/seguranca/Blocos

```
pera@Pera:~/facul/seguranca/Blocos$ cat cleartext.txt
This is our world now... the world of the electron and the switch, the
beauty of the baud.  We make use of a service already existing without paying
for what could be dirt-cheap if it wasn't run by profiteering gluttons, and
you call us criminals.  We explore... and you call us criminals.  We seek
after knowledge... and you call us criminals.  We exist without skin color,
without nationality, without religious bias... and you call us criminals.
You build atomic bombs, you wage wars, you murder, cheat, and lie to us
and try to make us believe it's for our own good, yet we're the criminals.
        Yes, I am a criminal.  My crime is that of curiosity.  My crime is
that of judging people by what they say and think, not what they look like.
My crime is that of outsmarting you, something that you will never forgive me
for.
        I am a hacker, and this is my manifesto.  You may stop this individual,
but you can't stop us all... after all, we're all alike.
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -aes-128-ecb -in cleartext.txt -K $(cat key.priv) -out ciphertext.tx
t
pera@Pera:~/facul/seguranca/Blocos$ cat ciphertext.txt
[illegible]
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -d -aes-128-ecb -in ciphertext.txt -K $(cat key.priv)
This is our world now... the world of the electron and the switch, the
beauty of the baud.  We make use of a service already existing without paying
for what could be dirt-cheap if it wasn't run by profiteering gluttons, and
you call us criminals.  We explore... and you call us criminals.  We seek
after knowledge... and you call us criminals.  We exist without skin color,
without nationality, without religious bias... and you call us criminals.
You build atomic bombs, you wage wars, you murder, cheat, and lie to us
and try to make us believe it's for our own good, yet we're the criminals.
        Yes, I am a criminal.  My crime is that of curiosity.  My crime is
that of judging people by what they say and think, not what they look like.
My crime is that of outsmarting you, something that you will never forgive me
for.
        I am a hacker, and this is my manifesto.  You may stop this individual,
but you can't stop us all... after all, we're all alike.
```

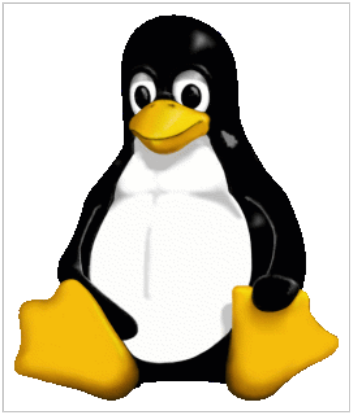
Cifrando o texto, o `ciphertext.txt` vira só lixo. Usando a mesma chave com `-d` o texto volta ao normal.

Prática 1: modo ECB com a imagem do Tux

```
pera@Pera: ~/facul/seguranca/Blocos
pera@Pera:~/facul/seguranca/Blocos$ wget https://upload.wikimedia.org/wikipedia/commons/a/af/Tux.png
--2026-10-20 03:06:15--  https://upload.wikimedia.org/wikipedia/commons/a/af/Tux.png
Resolving upload.wikimedia.org (upload.wikimedia.org)... 195.200.68.240, 2a02:ec80:700:ed1a::2:b
Connecting to upload.wikimedia.org (upload.wikimedia.org)|195.200.68.240|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11913 (12K) [image/png]
Saving to: 'Tux.png'

     0K .....                                                 100%  445M=0s

2026-10-20 03:06:15 (445 MB/s) - 'Tux.png' saved [11913/11913]
pera@Pera:~/facul/seguranca/Blocos$ convert Tux.png tux.ppm
pera@Pera:~/facul/seguranca/Blocos$ head -n 3 tux.ppm > header.txt
pera@Pera:~/facul/seguranca/Blocos$ tail -n +4 tux.ppm > body.bin
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -aes-128-ecb -in body.bin -K $(cat key.priv) -out body.ecb.enc
pera@Pera:~/facul/seguranca/Blocos$ cat header.txt body.ecb.enc > tux-ecb-enc.ppm
```



Original

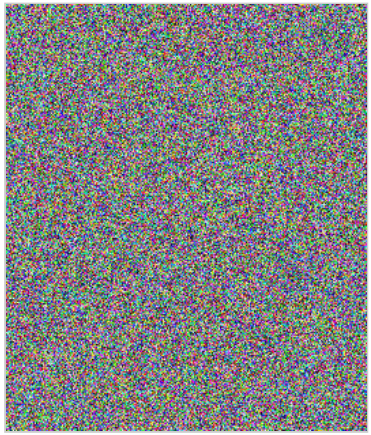


ECB

Mesmo cifrada dá pra ver o Tux. Isso acontece porque no ECB cada bloco é cifrado separado, então blocos iguais na imagem continuam iguais depois de cifrar, e as partes da mesma cor deixam o desenho aparecendo. Ou seja, o ECB vaza informação.

Prática 2: modo CBC

```
pera@Pera: ~/facul/seguranca/Blocos
pera@Pera:~/facul/seguranca/Blocos$ openssl rand -hex 16 > iv.txt
pera@Pera:~/facul/seguranca/Blocos$ cat iv.txt
b784d28faa639877b17cc77b1e189353
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -aes-128-cbc -in body.bin -K $(cat key.priv) -iv $(cat iv.txt) -out
body.cbc.enc
pera@Pera:~/facul/seguranca/Blocos$ cat header.txt body.cbc.enc > tux-cbc-enc.ppm
```

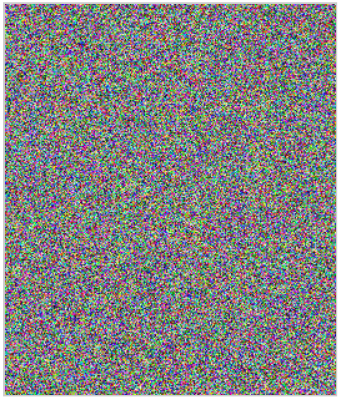


CBC

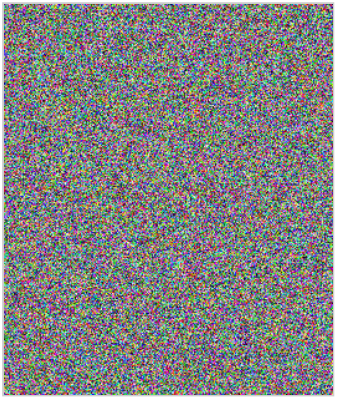
No CBC cada bloco depende do bloco anterior (e o primeiro depende do IV), então blocos iguais não ficam iguais. A imagem vira só ruído e não dá mais pra ver o Tux.

Testes com outros modos: CFB, OFB e CTR

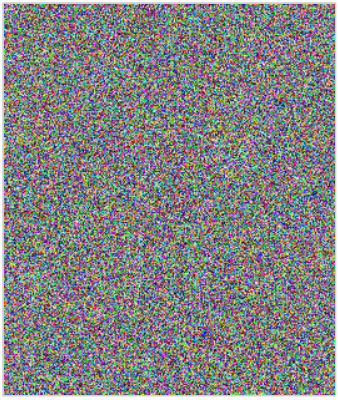
```
pera@Pera: ~/facul/seguranca/Blocos
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -aes-128-cfb -in body.bin -K $(cat key.priv) -iv $(cat iv.txt) -out
body.cfb.enc
pera@Pera:~/facul/seguranca/Blocos$ cat header.txt body.cfb.enc > tux-cfb-enc.ppm
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -aes-128-ofb -in body.bin -K $(cat key.priv) -iv $(cat iv.txt) -out
body.ofb.enc
pera@Pera:~/facul/seguranca/Blocos$ cat header.txt body.ofb.enc > tux-ofb-enc.ppm
pera@Pera:~/facul/seguranca/Blocos$ openssl enc -aes-128-ctr -in body.bin -K $(cat key.priv) -iv $(cat iv.txt) -out
body.ctr.enc
pera@Pera:~/facul/seguranca/Blocos$ cat header.txt body.ctr.enc > tux-ctr-enc.ppm
```



CFB



OFB



CTR

Nos modos CFB, OFB e CTR o resultado também foi ruído, igual ao CBC. Dá pra concluir que o problema não é o AES e sim o modo usado, e que o ECB não deve ser usado.